

Ex. 11. Simplify : $\frac{(6.25)^{\frac{1}{2}} \times (0.0144)^{\frac{1}{2}} + 1}{(0.027)^{\frac{1}{3}} \times (81)^{\frac{1}{4}}}$.

(S.S.C., 2005)

Sol. Given expression = $\frac{\{(2.5)^2\}^{\frac{1}{2}} \times \{(0.12)^2\}^{\frac{1}{2}} + 1}{\{(0.3)^3\}^{\frac{1}{3}} \times (3^4)^{\frac{1}{4}}} = \frac{(2.5)^{\left(2 \times \frac{1}{2}\right)} \times (0.12)^{\left(2 \times \frac{1}{2}\right)} + 1}{(0.3)^{\left(3 \times \frac{1}{3}\right)} \times 3^{\left(4 \times \frac{1}{4}\right)}}$

$$= \frac{2.5 \times 0.12 + 1}{0.3 \times 3} = \frac{0.3 + 1}{0.9} = \frac{1.3}{0.9} = \frac{13}{9} = 1.444 \dots = 1.\bar{4}.$$

Ex. 12. Find the value of $\frac{(243)^{\frac{n}{5}} \cdot 3^{2n+1}}{9^n \times 3^{n-1}}$.

Sol. $\frac{(243)^{\frac{n}{5}} \cdot 3^{2n+1}}{9^n \times 3^{n-1}} = \frac{(3^5)^{\frac{n}{5}} \times 3^{2n+1}}{(3^2)^n \times 3^{n-1}} = \frac{3^{\left(5 \times \frac{n}{5}\right)} \times 3^{2n+1}}{3^{2n} \times 3^{n-1}} = \frac{3^n \times 3^{2n+1}}{3^{2n} \times 3^{n-1}}$

$$= \frac{3^{n+(2n+1)}}{3^{2n+n-1}} = \frac{3^{(3n+1)}}{3^{(3n-1)}} = 3^{(3n+1)-(3n-1)} = 3^2 = 9.$$

Ex. 13. Find the value of $\left(2^{\frac{1}{4}} - 1\right) \left(2^{\frac{3}{4}} + 2^{\frac{1}{2}} + 2^{\frac{1}{4}} + 1\right)$.

(N.I.F.T., 2003)

Sol. Putting $2^{\frac{1}{4}} = x$, we get :

$$\begin{aligned} \left(2^{\frac{1}{4}} - 1\right) \left(2^{\frac{3}{4}} + 2^{\frac{1}{2}} + 2^{\frac{1}{4}} + 1\right) &= (x - 1)(x^3 + x^2 + x + 1) \\ &= (x - 1)[x^2(x + 1) + (x + 1)] \\ &= (x - 1)(x + 1)(x^2 + 1) = (x^2 - 1)(x^2 + 1) \\ &= (x^4 - 1) = \left[\left(2^{\frac{1}{4}}\right)^4 - 1\right] = \left[2^{\left(\frac{1}{4} \times 4\right)} - 1\right] = (2 - 1) = 1. \end{aligned}$$

Ex. 14. Find the value of $\frac{6^{\frac{2}{3}} \times \sqrt[3]{6^7}}{\sqrt[3]{6^6}}$.

Sol. $\frac{6^{\frac{2}{3}} \times \sqrt[3]{6^7}}{\sqrt[3]{6^6}} = \frac{6^{\frac{2}{3}} \times (6^7)^{\frac{1}{3}}}{(6^6)^{\frac{1}{3}}} = \frac{6^{\frac{2}{3}} \times 6^{\left(7 \times \frac{1}{3}\right)}}{6^{\left(6 \times \frac{1}{3}\right)}} = \frac{6^{\frac{2}{3}} \times 6^{\left(\frac{7}{3}\right)}}{6^2}$

$$= 6^{\frac{2}{3}} \times 6^{\left(\frac{7}{3} - 2\right)} = 6^{\frac{2}{3}} \times 6^{\frac{1}{3}} = 6^{\left(\frac{2}{3} + \frac{1}{3}\right)} = 6^1 = 6.$$

Ex. 15. If $\left(\frac{p}{q}\right)^{rx-s} = \left(\frac{q}{p}\right)^{px-q}$, then find the value of x .

Sol. $\left(\frac{p}{q}\right)^{rx-s} = \left(\frac{q}{p}\right)^{px-q} \Leftrightarrow \left(\frac{p}{q}\right)^{rx-s} = \left(\frac{p}{q}\right)^{-(px-q)}$

$$\begin{aligned} \Leftrightarrow rx - s &= -(px - q) \Leftrightarrow rx - s = -px + q \\ \Leftrightarrow rx + px &= q + s \Leftrightarrow x(p + r) = q + s \\ \Leftrightarrow x &= \frac{q + s}{p + r}. \end{aligned}$$